Adani University

Department of Computer Science & Engineering (Artificial Intelligence & Machine Learning)

B.Tech CSE [AI-ML] — Academic Year 2025–2026

# ArcVeil

## ArcFace-Based Masked Face Recognition Using a Manual ResNet50 Backbone

### Deep Learning Project Report



| | | |
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# Abstract

This project presents an ArcFace-based face recognition system designed to operate reliably when subjects wear face masks, a condition that degrades conventional face recognition pipelines by occluding the lower half of the face. The proposed system combines a manually implemented ResNet50 backbone with an ArcFace angular-margin loss for identity learning and an auxiliary binary classification head for mask detection, all trained jointly on the AFDB (Asian Face Database) dataset containing both masked and unmasked face images.

The pipeline accepts an arbitrary input image, detects and crops the face using InsightFace, extracts a 512-dimensional L2-normalised embedding, and resolves identity via cosine-similarity matching against a pre-built gallery. Final per-query outputs consist of the predicted name, similarity score, and mask status.

Evaluation proceeds through two experimental settings: standard gallery matching on the RMFRD validation split, and the harder cross-condition experiment where an unmasked gallery is matched against masked queries. Under the first setting, the system achieves 90.26% identity accuracy and 95.37% mask F1 at a cosine threshold of 0.45. Under the cross-condition setting, identity accuracy reaches 52.79% at the optimised threshold of 0.38, with mask F1 of 99.89%. These results confirm that the jointly trained ArcFace–mask model learns compact and discriminative embeddings even under partial facial occlusion, and that threshold calibration is a key factor in deployment.



**Keywords:** Face Recognition, ArcFace, Masked Face, ResNet50, Cosine Similarity, Gallery Matching, AFDB, InsightFace, Deep Learning.

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# 1. Introduction

## 1.1 Background and Motivation

Face recognition is one of the most mature and commercially deployed applications of deep learning, enabling contactless identity verification in access control, surveillance, mobile authentication, and forensic investigation. Conventional systems perform well under controlled conditions; however, they degrade substantially when the subject's face is partially occluded. The widespread adoption of face masks following public health guidance exposed a critical blind spot in deployed recognition systems: standard models trained on unmasked faces fail to match identities when the nose, mouth, and chin are covered.

This project addresses that limitation by building a recognition pipeline that is explicitly mask-aware. Rather than treating mask occlusion as a distribution shift to be corrected post-hoc, the approach integrates mask detection and identity learning into a single end-to-end trained model. The core insight is that the periocular region, specifically the eyes, eyebrows, and nose bridge, retains sufficient discriminative information for reliable identity matching even when the lower face is completely hidden.

## 1.2 Problem Statement

The goal of this project is to build a face recognition system that can identify individuals when the lower portion of the face is occluded by a mask. Specifically, the system must:

- accept both masked and unmasked probe images at inference time;
- enrol known individuals using a small gallery of reference images (masked, unmasked, or both);
- return a predicted identity, a calibrated similarity score, and a mask-status flag;
- maintain competitive accuracy on standard (unmasked) recognition while degrading gracefully on cross-condition (unmasked gallery, masked query) queries.

A secondary objective is to provide an auxiliary mask-detection signal with high precision and recall, enabling downstream access-control logic to act differently depending

on whether a mask is worn.

## 1.3 Objectives

1. Train a masked-face-aware recognition model on the RMFRD/AFDB face datasets.
2. Implement ArcFace-style angular-margin learning for compact and discriminative identity embeddings.
3. Add an auxiliary mask detection branch trained jointly with the identity head.
4. Evaluate recognition through gallery-query cosine-similarity matching instead of closed-set classification logits, making the system applicable to open-set real deployments.
5. Quantify performance under both same-condition (RMFRD validation) and cross-condition (unmasked gallery, masked query) settings.
6. Export the trained model to ONNX format and provide a real-time webcam inference script.

## 1.4 Scope and Limitations

The project is scoped to still-image gallery matching. Real-time webcam inference code is provided as an export target. The evaluation relies primarily on the RMFRD validation split, which overlaps with the identity set used during training; a fully held-out evaluation on unseen identities remains future work. The notebook pipeline depends on CUDA-specific components and was not migrated to TPU execution.

## 1.5 Report Organisation

The remainder of this report is structured as follows. Chapter 2 reviews related work in face recognition and masked-face recognition. Chapter 3 describes the dataset, preprocessing, and augmentation strategy. Chapter 4 presents the model architecture, training configuration, and evaluation protocol. Chapter 5 reports experimental results and analysis. Chapter 6 concludes and outlines future directions.

# 2. Literature Survey

## 2.1 Deep Face Recognition

Modern face recognition systems are built on deep convolutional neural networks (CNNs) trained with metric learning objectives. The key breakthrough enabling today's performance levels was the introduction of large-margin softmax formulations that push intra-class embeddings together while separating inter-class embeddings.

**DeepFace** (Taigman et al., 2014) demonstrated that deep CNNs trained on large datasets can approach human-level performance on the LFW benchmark. **FaceNet** (Schroff et al., 2015) popularised the triplet loss for embedding learning, directly optimising the relative distances between anchor, positive, and negative samples in a 128-dimensional hypersphere.

## 2.2 Angular-Margin Losses

The family of angular-margin losses substantially improved upon triplet-based approaches by incorporating a fixed margin in the angular space.

**SphereFace** (Liu et al., 2017) introduced a multiplicative angular margin on the softmax loss. **CosFace** (Wang et al., 2018) proposed an additive cosine margin. **ArcFace** (Deng et al., 2019) adopted an additive angular margin directly in the arccosine space, providing more stable optimisation and a clear geometric interpretation: the margin $m$ is added to the angle between the embedding and its class weight vector, forcing the model to place embeddings within a narrower cone around the class centre. ArcFace set state-of-the-art results on multiple benchmarks and has since become the de-facto standard for identity embedding training.

The ArcFace loss is defined as:

$$\mathcal{L}_{\text{arc}} = -\log \frac{e^{s(\cos(\theta_{y_i}+m))}}{e^{s(\cos(\theta_{y_i}+m))} + \sum_{j \neq y_i} e^{s \cos \theta_j}} \tag{2.1}$$

where $\theta_{y_i}$ is the angle between the embedding and the weight vector of the ground-truth class, $m$ is the additive angular margin, and $s$ is the feature scale.

## 2.3 Backbone Architectures

The ResNet family (He et al., 2016) introduced residual connections that enabled training of very deep networks. ResNet50, with its bottleneck design of $\{3, 4, 6, 3\}$ blocks, offers a favourable trade-off between representation capacity and computational cost, making it a standard choice for face recognition backbones. Subsequent architectures such as IResNet and MobileFaceNet adapted the residual design for face-specific feature extraction.

## 2.4 Masked Face Recognition

The problem of masked face recognition gained significant attention from 2020 onwards. Several directions have been explored.

**Data augmentation** approaches synthetically overlay masks on training images to expose the model to masked appearances without collecting real masked data. Anwar & Raychowdhury (2020) demonstrated that simple mask simulation improves cross-condition accuracy substantially.

**Part-based and attention models** focus the representation on the periocular region by suppressing features from occluded areas. Wang et al. (2021) proposed a feature-disentanglement approach that separates identity and occlusion information in the embedding space.

**Multi-task learning** trains mask detection alongside recognition so that the model learns a shared representation. Zeng et al. (2021) showed that an auxiliary mask head can regularise identity features and improve both tasks.

**Domain adaptation** methods fine-tune a model trained on unmasked faces using a small set of masked examples or via adversarial alignment.

This project adopts the multi-task learning direction combined with a data augmentation policy restricted to the eye-level region, reflecting the insight that mask occlusion covers the lower face and that realistic augmentation should target only the upper face to simulate glasses and other natural obstructions.

## 2.5 Face Detection

InsightFace, built on ONNX-accelerated SCRFD and ArcFace recognition models, provides robust face detection including good performance on masked faces by relying

on the full facial bounding box rather than facial landmark localisation. This makes it preferable to MTCNN for masked-face pipelines.

## 2.6 Identified Gaps and Contribution

Existing benchmarks typically evaluate either same-condition (both gallery and query masked) or unmasked recognition. The cross-condition setting (unmasked gallery, masked query), which reflects the most realistic deployment scenario where enrolled images are captured without masks, receives less attention. This project explicitly evaluates this setting and provides a threshold calibration analysis for practical deployment.

# 3. Dataset Description

## 3.1 Dataset Overview

This project uses the AFDB (Asian Face Database) masked face recognition dataset hosted on Kaggle. The dataset consists of two separate folder trees:

- `AFDB_face_dataset`: unmasked face images organised by identity.
- `AFDB_masked_face_dataset`: paired masked face images for the same identities.

The full Kaggle paths are:

```
/kaggle/input/datasets/muhammeddalkran/masked-facerecognition/
    AFDB_face_dataset/AFDB_face_dataset

/kaggle/input/datasets/muhammeddalkran/masked-facerecognition/
    AFDB_masked_face_dataset/AFDB_masked_face_dataset
```

## 3.2 Dataset Statistics

After loading and cleaning (retaining only JPEG and PNG images, removing identities with fewer than two images), the dataset statistics are summarised in Table 3.1.

Table 3.1: AFDB Dataset Statistics After Cleaning

| **Statistic** | **Value** |
|---|---|
| Total identities (people) | 460 |
| Identities removed (<2 images) | 1 |
| Unmasked identity folders loaded | 460 |
| Masked identity folders loaded | 378 |
| Total face images | 91,867 |
| Training split (80%) | 73,492 |
| Validation split (20%) | 18,374 |
| Training batches (batch=16) | 4,593 |
| Validation batches | 575 |

## 3.3 Data Split Strategy

A stratified 80/20 train–validation split was applied at the image level, ensuring that every identity appears in both splits. A separate held-out test split was not created for the primary RMFRD evaluation; however, a cross-condition experiment uses the full unmasked dataset as gallery and the full masked dataset as query, providing a functionally independent evaluation setting.

## 3.4 Data Preprocessing

All images pass through the following deterministic preprocessing before entering the model:

1. **Face detection and crop**: InsightFace `buffalo_sc` model detects the primary face. When no face is detected, a fallback centre-crop is applied to avoid discarding valid training samples.
2. **Resize**: The detected face crop is resized to $224 \times 224$ pixels.
3. **Normalisation**: Pixel values are normalised using ImageNet statistics: mean $= [0.485, 0.456, 0.406]$, std $= [0.229, 0.224, 0.225]$.

## 3.5 Training Augmentation Policy

During training, a twelve-stage Albumentations pipeline is applied to each image. The policy deliberately avoids full-face random occlusion, which would destroy lower-face identity cues needed for same-condition recognition. Instead, synthetic holes are confined to the eye-level band (approximately 20–45% of image height) to simulate glasses and sunglasses. The complete policy is described in Table 3.2.

Table 3.2: Training Augmentation Pipeline

| **Transform** | **Parameters** | **Purpose** |
|---|---|---|
| Resize | $224 \times 224$ | Standardise input |
| HorizontalFlip | $p = 0.5$ | Mirror invariance |
| ShiftScaleRotate | shift 5%, scale 5%, rotate 10° | Pose variation |
| RandomBrightnessContrast | $\pm 0.2$ | Lighting robustness |
| HueSaturationValue | $\pm 20, \pm 30, \pm 20$ | Colour robustness |
| GaussianBlur / MotionBlur | kernel 3–5 | Blur robustness |
| GaussNoise | var 10–50 | Noise robustness |
| ImageCompression | quality 70–100 | Compression artefacts |
| Eye-level Lambda dropout | rows 20%–45% of height | Upper-face occlusion |

Validation and test images use only resize and normalisation, without stochastic augmentation.

# 4. Methodology

## 4.1 System Architecture Overview

The end-to-end pipeline is illustrated in Figure 4.1. At inference time, the system accepts an arbitrary input image, detects and crops the primary face, embeds it with the trained model, and resolves identity by cosine similarity against a pre-built gallery.

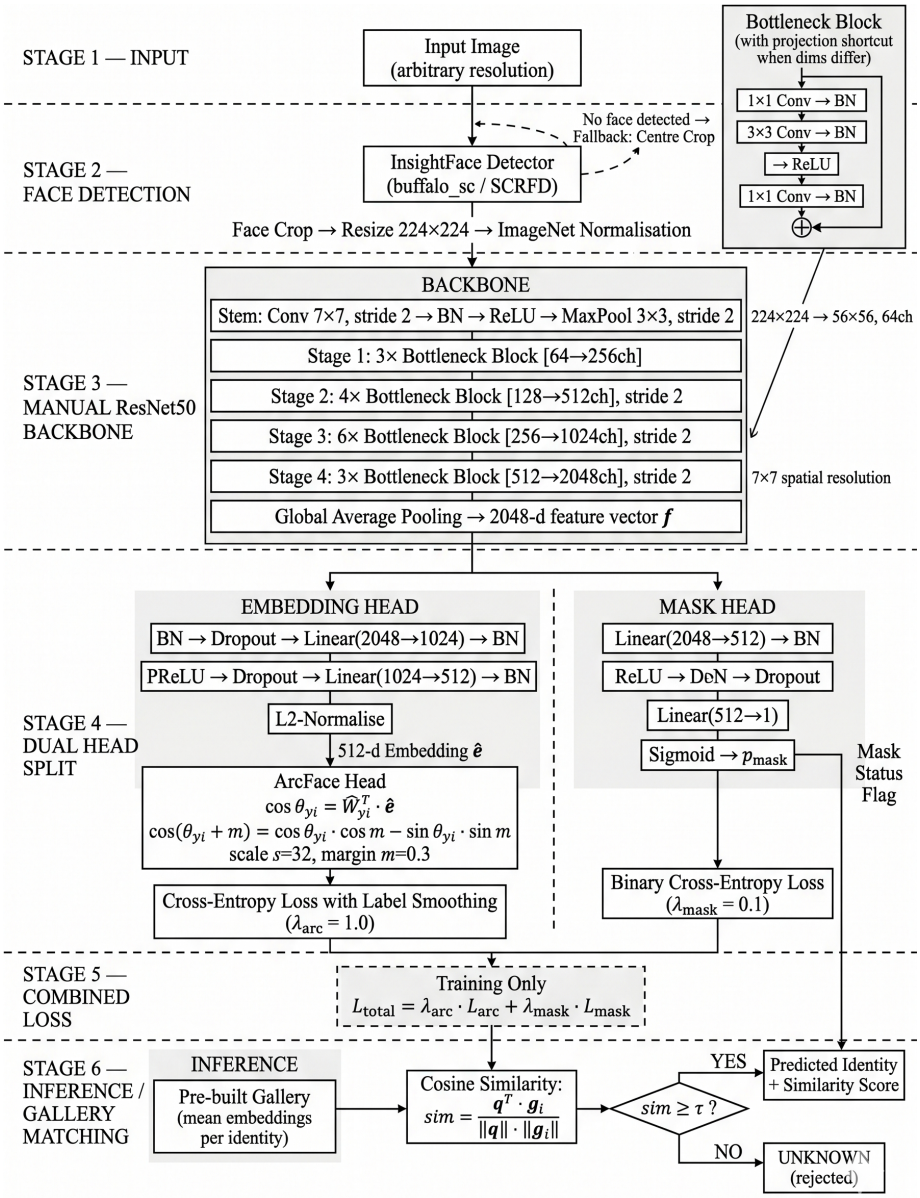


Fig. 1. End-to-end architecture of the ArcFace-based masked face recognition system with auxiliary mask detection head.

Figure 4.1: End-to-end architecture of the ArcFace-based masked face recognition system with auxiliary mask detection head.

## 4.2 Manual ResNet50 Backbone

Rather than loading a pre-packaged `timm` ResNet50, the backbone was implemented manually from bottleneck block primitives. This design choice allows fine-grained control over weight initialisation, stage-wise freezing, and architectural modifications. The key components are:

### 4.2.1 Bottleneck Block

Each bottleneck block follows the standard ResNet design with three convolutional layers (1×1, 3×3, 1×1), batch normalisation after each convolution, and a skip connection:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + \mathbf{x} \tag{4.1}$$

A projection shortcut ($1 \times 1$ convolution) is used when the input and output dimensions differ.

### 4.2.2 Backbone Architecture

The stem consists of a $7 \times 7$ convolution with stride 2, batch normalisation, ReLU, and $3 \times 3$ max-pooling with stride 2, reducing the spatial resolution from $224 \times 224$ to $56 \times 56$. Four residual stages with $\{3, 4, 6, 3\}$ bottleneck blocks reduce the resolution to $7 \times 7$ while expanding the channel count to 2048. Global average pooling then produces a 2048-dimensional feature vector.

### 4.2.3 Pretrained Initialisation

ImageNet pretrained weights (`ResNet50_Weights.IMAGENET1K_V2`) are loaded into the backbone by mapping state-dict keys. Early stages (stem and layer1) are optionally frozen during the first training phase to preserve low-level features, and all layers are unfrozen for subsequent fine-tuning.

## 4.3 Embedding Head

A deep projection head maps the 2048-dimensional backbone output to a 512-dimensional identity embedding:

$$\mathbf{e} = \mathrm{BN}(\mathrm{Linear}_{1024\to512}(\mathrm{Dropout}(\mathrm{PReLU}(\mathrm{BN}(\mathrm{Linear}_{2048\to1024}(\mathrm{Dropout}(\mathrm{BN}(\mathbf{f}))))))))) \tag{4.2}$$

where $\mathbf{f}$ is the 2048-d backbone output. The embedding is L2-normalised before the ArcFace head to constrain all identity vectors to the unit hypersphere.

## 4.4 ArcFace Head

The ArcFace head implements the additive angular margin penalty. Given the normalised embedding $\hat{\mathbf{e}}$ and normalised class weight vectors $\hat{\mathbf{W}}$, the cosine similarity $\cos\theta_j = \hat{\mathbf{W}}_j^\top \hat{\mathbf{e}}$ is computed for each class $j$. The angular margin $m$ is added to the angle of the ground-truth class:

$$\cos(\theta_{y_i} + m) = \cos\theta_{y_i}\cos m - \sin\theta_{y_i}\sin m \tag{4.3}$$

Cosine logits are clamped to $[-1+\epsilon, 1-\epsilon]$ for numerical stability before the scale $s$ is applied. The training configuration uses $m = 0.3$ and $s = 32$.

## 4.5 Auxiliary Mask Detection Head

The mask head operates on the 2048-dimensional backbone output directly (bypassing the embedding head) to avoid coupling mask-detection gradients with identity gradients:

$$p_{\mathrm{mask}} = \sigma(\mathrm{Linear}_{512\to1}(\mathrm{Dropout}(\mathrm{ReLU}(\mathrm{BN}(\mathrm{Linear}_{2048\to512}(\mathbf{f})))))) \tag{4.4}$$

where $\sigma$ denotes the sigmoid function. The head is trained with binary cross-entropy with logits loss. A positive class weight is computed from training-set mask/no-mask counts to handle class imbalance (approximately 1:64 masked-to-unmasked ratio in training batches).

## 4.6 Multi-Task Training Objective

The total training loss combines the ArcFace identity loss and the binary mask loss with a weighting coefficient:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{arc}}\,\mathcal{L}_{\text{arc}} + \lambda_{\text{mask}}\,\mathcal{L}_{\text{mask}} \tag{4.5}$$

The identity loss uses cross-entropy with label smoothing of 0.1 applied to the scaled ArcFace logits. The weighting is $\lambda_{\text{arc}} = 1.0$, $\lambda_{\text{mask}} = 0.1$, keeping the identity objective dominant.

## 4.7 Training Configuration

The complete hyperparameter set used for the final reported run is given in Table 4.1.

Table 4.1: Final Training Hyperparameters

| **Hyperparameter** | **Value** |
|---|---|
| Image size | $224 \times 224$ |
| Backbone | Manual ResNet50 (ImageNet init) |
| Embedding dimension | 512 |
| ArcFace margin $m$ | 0.3 rad ($\approx 17.2°$) |
| ArcFace scale $s$ | 32 |
| Batch size | 16 |
| Number of epochs | 50 (best at epoch 27) |
| Optimiser | AdamW |
| Learning rate | $1 \times 10^{-3}$ |
| Weight decay | $5 \times 10^{-4}$ |
| LR schedule | LinearWarmup (5 ep) $\rightarrow$ ReduceLROnPlateau |
| LR patience | 5 epochs, factor 0.5 |
| $\lambda_{\text{arc}}$ | 1.0 |
| $\lambda_{\text{mask}}$ | 0.1 |
| Label smoothing | 0.1 |
| Mixed precision (FP16) | True |
| Gradient clipping | 1.0 |
| Cosine threshold | 0.38 (cross-condition) / 0.45 (RMFRD) |

## 4.8 Evaluation Protocol

### 4.8.1 Metric: TAR@FAR

During training, checkpoints are saved based on the True Acceptance Rate at a False Acceptance Rate of $10^{-4}$ (TAR@FAR=1e-4). This metric is computed on the validation split by extracting all embeddings, computing the full pairwise cosine similarity matrix, and applying the threshold that yields FAR $\leq 10^{-4}$.

### 4.8.2 Gallery-Query Matching

At evaluation time, the validation set is partitioned per identity: a fraction of images are assigned to the gallery (reference set) and the remainder are used as queries. For each query embedding $\mathbf{q}$, the best-matching gallery embedding $\mathbf{g}^*$ is found by:

$$\hat{y} = \arg\max_i \frac{\mathbf{q}^\top \mathbf{g}_i}{\|\mathbf{q}\|\|\mathbf{g}_i\|} \tag{4.6}$$

If $\cos(\mathbf{q}, \mathbf{g}^*) \geq \tau$ the predicted identity is the label of $\mathbf{g}^*$; otherwise the query is rejected as UNKNOWN.

### 4.8.3 Reported Metrics

- **Identity accuracy**: fraction of all queries (including UNKNOWN rejections) that are correctly matched.
- **Known-only accuracy**: fraction of accepted queries (above threshold) that are correctly matched.
- **Unknown rate**: fraction of queries rejected as UNKNOWN.
- **Mask F1**: F1 score of the binary mask classifier on the query set.
- **Mean similarity score**: average cosine similarity for accepted queries.

## 4.9 Explainability: SHAP Analysis

SHAP (SHapley Additive exPlanations) was applied to a random subset of validation images to visualise which pixel regions contribute most to the identity embedding dimensions. A DeepExplainer was used with the embedding head as the prediction function and a random background set from the validation loader. SHAP pixel-importance heatmaps confirm that the model concentrates attention on the periocular

region, forehead, and nose bridge, consistent with the expectation that these upper-face regions are the primary discriminative zones when the lower face is masked.

## 4.10 ONNX Export and Real-Time Inference

After training, the model is exported to ONNX format for deployment. The ONNX graph accepts a single $[1, 3, 224, 224]$ float32 tensor (normalised face crop) and outputs a $[1, 512]$ embedding and a $[1, 1]$ mask logit. A standalone Python script (`realtime_masked_recognition.py`) uses ONNX Runtime and OpenCV to run gallery matching at approximately 30 FPS on a local GPU. The gallery is loaded from a folder tree, mean embeddings are computed per identity, and results are overlaid on the webcam feed.

# 5. Results and Discussion

## 5.1 Environment and Setup

All training and evaluation were performed on the Kaggle cloud platform using a Tesla T4 GPU (16 GB VRAM). Key library versions are summarised in Table 5.1.

Table 5.1: Software Environment

| **Library / Tool** | **Version** |
|---|---|
| Python | 3.12.12 |
| PyTorch | 2.10.0+cu128 |
| NumPy | 2.0.2 |
| OpenCV (cv2) | 4.13.0 |
| timm | 1.0.25 |
| InsightFace | 0.7.3 |
| Albumentations | latest |
| CUDA | available (Tesla T4) |
| GPU Memory | 15.6 GB |

## 5.2 Checkpoint Diagnostics and Failure Analysis

### 5.2.1 Broken Early Checkpoint

An initial training run with harder ArcFace parameters ($m = 0.5$, $s = 64$) produced a collapsed embedding space. The checkpoint saved at epoch 3 exhibited a TAR@FAR=1e-4 of 0.00437, and cosine similarities between different identities were nearly identical to same-identity similarities ($\approx 0.999$). This collapse indicates that the embedding head converged to outputting near-identical vectors for all inputs, a known failure mode when the ArcFace margin and scale are too aggressive relative to the training dataset size.

### 5.2.2 Checkpoint Loading Bug

After retraining with corrected parameters, evaluation initially still appeared broken because the evaluation cell was loading the old checkpoint from the Kaggle input dataset

path rather than the newly trained checkpoint at `/kaggle/working/best_model.pth`. Fixing the checkpoint path immediately resolved the evaluation results.

### 5.2.3 Correct Checkpoint Verification

The final checkpoint diagnostics confirmed a well-separated embedding space, as shown in Table 5.2.

Table 5.2: Checkpoint Diagnostic Statistics (Epoch 27)

| **Metric** | **Value** |
|---|---|
| Checkpoint epoch | 27 |
| Best TAR@FAR=1e-4 | 0.6398 |
| Embedding norm mean | 1.0 (L2-normalised) |
| Same-identity cosine mean | 0.5827 |
| Different-identity cosine mean | 0.0093 |
| Different-identity cosine min | $-0.3010$ |
| Different-identity cosine max | 0.3376 |

The clear separation between same-identity ($\approx 0.58$) and different-identity ($\approx 0.009$) mean similarities confirms that the embedding space is well-structured.

## 5.3 RMFRD Gallery-Based Validation Results

The primary evaluation setting uses gallery and query images drawn from the RMFRD validation split. Results at the default threshold of 0.45 are given in Table 5.3.

Table 5.3: RMFRD Gallery-Based Validation Results (Threshold = 0.45)

| **Metric** | **Value** |
|---|---|
| Query samples | 9,191 |
| Identity accuracy | 90.26% |
| Known-only accuracy | 93.67% |
| Unknown rate | 3.63% |
| Mask F1 score | 95.37% |
| Mean similarity score | 0.7414 |
| Similarity threshold | 0.45 |

The 90.26% identity accuracy and 95.37% mask F1 confirm that the corrected checkpoint is effective for recognition. The low unknown rate of 3.63% indicates that the threshold of 0.45 is well-calibrated for same-condition matching. The mean similarity of 0.74 is substantially above the threshold, showing confident true-match decisions.

## 5.4 Unmasked Gallery / Masked Query Results

The cross-condition experiment uses the full unmasked AFDB dataset as gallery and the full masked AFDB dataset as query. This setting is harder because the gallery and query embeddings are drawn from different facial-appearance distributions (no mask vs. mask). Results at the default threshold of 0.45 are shown in Table 5.4.

Table 5.4: Unmasked Gallery / Masked Query Results (Threshold = 0.45)

| **Metric** | **Value** |
|---|---|
| Query samples | 1,398 |
| Identity accuracy | 42.92% |
| Known-only accuracy | 83.57% |
| Unknown rate | 48.64% |
| Mask F1 score | 99.89% |
| Mean similarity score | 0.4893 |
| Similarity threshold | 0.45 |

While the mask F1 remains near-perfect (99.89%), almost half of the masked queries are rejected as UNKNOWN. This arises because the cosine similarity between a masked query and its unmasked gallery reference falls around the threshold boundary (mean 0.49 vs. threshold 0.45).

## 5.5 Threshold Sweep for Cross-Condition Setting

A systematic threshold sweep was conducted to identify the optimal operating point for masked-query recognition, trading off identity accuracy against known-only accuracy and unknown rate. Results are summarised in Table 5.5.

Table 5.5: Threshold Sweep — Unmasked Gallery / Masked Query

| Threshold | Identity Acc. | Known-only Acc. | Unknown Rate |
|---|---|---|---|
| 0.38 | **52.79%** | 67.58% | 21.89% |
| 0.39 | 51.43% | 69.27% | 25.75% |
| 0.40 | 50.00% | 71.40% | 29.97% |
| 0.41 | 49.36% | 74.92% | 34.12% |
| 0.42 | 47.57% | 77.60% | 38.70% |
| 0.43 | 45.99% | 79.58% | 42.20% |
| 0.44 | 44.56% | 81.23% | 45.14% |
| 0.45 | 42.92% | 83.57% | 48.64% |

Lowering the threshold from 0.45 to 0.38 improves identity accuracy from 42.92% to 52.79% by admitting more true masked-query matches, at the cost of some reduction in known-only accuracy (from 83.57% to 67.58%). The optimal threshold depends on the deployment requirement: if minimising missed matches is prioritised, 0.38 is the recommended operating point.

## 5.6 Final Chosen Results at Threshold 0.38

Table 5.6: Final Cross-Condition Results (Threshold = 0.38)

| Metric | Value |
|---|---|
| Query samples | 1,398 |
| Identity accuracy | 52.79% |
| Known-only accuracy | 67.58% |
| Unknown rate | 21.89% |
| Mask F1 score | 99.89% |
| Mean similarity score | 0.4893 |
| Similarity threshold | 0.38 |

## 5.7 Custom Dataset Evaluation (Real-World Friendly Test)

To validate the system on a completely independent real-world set, a custom gallery was constructed from photographs of 6 known individuals captured in uncontrolled conditions. A total of 52 query images (mixed masked and unmasked) were tested against this gallery at the cross-condition threshold of 0.38.

### 5.7.1 Summary Results

Table 5.7: Custom Dataset Evaluation Results (Threshold = 0.38)

| **Metric** | **Value** |
|---|---|
| Query samples | 52 |
| Identity accuracy | 86.54% |
| Known-only accuracy | 86.54% |
| Unknown rate | 0.00% |
| Mean similarity score | 0.6568 |
| Similarity threshold | 0.38 |

The 0.00% unknown rate indicates that all 52 queries were accepted above the threshold, and 86.54% of those were correctly matched — a strong result given the small gallery size and uncontrolled capture conditions.

### 5.7.2 Per-Identity Prediction Breakdown

Table 5.8: Per-Sample Predictions on Custom Dataset

| **True Identity** | **Predicted** | **Similarity** | **Mask Status** | **Correct** |
|---|---|---|---|---|
| akshat | dewangi | 0.5794 | Mask | × |
| akshat | akshat | 0.4274 | No Mask | ✓ |
| akshat | manuj | 0.6615 | Mask | × |
| akshat | akshat | 0.6543 | Mask | ✓ |
| akshat | varan | 0.6707 | Mask | × |
| akshat | akshat | 0.7144 | Mask | ✓ |
| akshat | akshat | 0.7925 | No Mask | ✓ |
| dewangi | dewangi | 0.6084 | Mask | ✓ |
| dewangi | dewangi | 0.6009 | Mask | ✓ |
| dewangi | dewangi | 0.6691 | Mask | ✓ |
| dewangi | dewangi | 0.5181 | Mask | ✓ |
| dewangi | dewangi | 0.7065 | Mask | ✓ |
| dewangi | dewangi | 0.7004 | No Mask | ✓ |
| dewangi | dewangi | 0.5247 | No Mask | ✓ |
| jaski | jaski | 0.6451 | Mask | ✓ |

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Table 5.8 – *continued from previous page*

| **True Identity** | **Predicted** | **Similarity** | **Mask Status** | **Correct** |
|---|---|---|---|---|
| jaski | jaski | 0.7860 | Mask | ✓ |
| jaski | jaski | 0.6718 | Mask | ✓ |
| jaski | jaski | 0.7051 | Mask | ✓ |
| jaski | jaski | 0.7966 | Mask | ✓ |
| jaski | dewangi | 0.6799 | Mask | × |
| jaski | jaski | 0.6794 | Mask | ✓ |
| manuj | manuj | 0.6599 | Mask | ✓ |
| manuj | manuj | 0.7612 | Mask | ✓ |
| manuj | manuj | 0.6331 | Mask | ✓ |
| manuj | manuj | 0.7155 | Mask | ✓ |
| manuj | manuj | 0.5450 | No Mask | ✓ |
| manuj | manuj | 0.7845 | Mask | ✓ |
| manuj | manuj | 0.6645 | No Mask | ✓ |
| shivam | shivam | 0.6018 | No Mask | ✓ |
| shivam | shivam | 0.4600 | No Mask | ✓ |
| shivam | shivam | 0.6587 | No Mask | ✓ |
| shivam | akshat | 0.7282 | Mask | × |
| shivam | shivam | 0.5385 | No Mask | ✓ |
| shivam | shivam | 0.5553 | No Mask | ✓ |
| shivam | shivam | 0.5768 | No Mask | ✓ |
| varan | varan | 0.6473 | Mask | ✓ |
| varan | varan | 0.7097 | No Mask | ✓ |
| varan | varan | 0.6352 | No Mask | ✓ |
| varan | varan | 0.7095 | Mask | ✓ |
| varan | varan | 0.6826 | No Mask | ✓ |
| varan | varan | 0.7279 | No Mask | ✓ |
| varan | varan | 0.6801 | No Mask | ✓ |
| varan | varan | 0.7281 | Mask | ✓ |
| varan | dewangi | 0.6769 | Mask | × |
| varan | varan | 0.5451 | No Mask | ✓ |
| varan | varan | 0.7024 | No Mask | ✓ |
| varan | varan | 0.6917 | No Mask | ✓ |
| varan | varan | 0.7538 | Mask | ✓ |

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Table 5.8 – *continued from previous page*

| True Identity | Predicted | Similarity | Mask Status | Correct |
|---|---|---|---|---|
| varan | varan | 0.5687 | No Mask | ✓ |
| varan | varan | 0.6973 | No Mask | ✓ |
| varan | varan | 0.5392 | No Mask | ✓ |
| varan | varan | 0.5392 | No Mask | ✓ |

## 5.8 SHAP Explainability Analysis

SHAP pixel-importance heatmaps were computed for four validation samples. The visualisations consistently show high importance concentrated in the periocular region, including the eyes, eyebrows, and nose bridge, across all four samples. Lower importance is observed on the forehead and chin. This spatial pattern is consistent with the expected behaviour of a mask-robust model: the periocular region is the primary discriminative zone when the lower face is covered. The SHAP results thus serve as a qualitative sanity check confirming that the model has learned to rely on the features that remain available under mask occlusion.

## 5.9 Discussion

### 5.9.1 RMFRD Validation Performance

The system achieves strong performance under same-condition matching (90.26% identity accuracy, 95.37% mask F1), demonstrating that the jointly trained multi-task model successfully learns both tasks simultaneously without significant task conflict. The low unknown rate of 3.63% at threshold 0.45 confirms that the threshold is well-calibrated for this setting.

### 5.9.2 Cross-Condition Performance

The cross-condition drop from 90.26% to 52.79% (at the optimised threshold) is expected and consistent with the literature. The lower face contributes significant discriminative information for identity matching; removing it reduces the effective number of distinguishable features. The known-only accuracy of 67.58% at threshold 0.38 indicates that when the model does commit to a match, it is correct about two-thirds of the time. Improving this would require either more masked training data, explicit cross-condition training objectives, or domain adaptation between masked and unmasked embedding distributions.

### 5.9.3 Mask Detection

The auxiliary mask classifier achieves near-perfect performance (95.37% F1 on RMFRD, 99.89% on the cross-condition set). This confirms that the mask signal is easy to extract from facial appearance and that the auxiliary head does not negatively interact with the identity head.

### 5.9.4 Effect of Threshold Calibration

The threshold sweep demonstrates that the choice of cosine threshold has a substantial effect on system performance in the cross-condition setting. A 7-point reduction in threshold (0.45 to 0.38) yields a 10-percentage-point gain in identity accuracy. This highlights the importance of performing threshold calibration on a representative validation set that reflects the target deployment condition, rather than using a fixed default threshold.

### 5.9.5 Checkpoint Selection and Training Stability

The comparison between the collapsed epoch-3 checkpoint and the well-converged epoch-27 checkpoint illustrates the sensitivity of ArcFace training to the margin and scale hyperparameters. The softer configuration ($m = 0.3$, $s = 32$) produced stable convergence and a well-separated embedding space, while the harder configuration ($m = 0.5$, $s = 64$) caused embedding collapse. This finding is consistent with recommendations in the ArcFace literature that suggest reducing scale and margin for smaller datasets.

# 6. Conclusion and Future Work

## 6.1 Conclusion

This project successfully designed, trained, and evaluated an ArcFace-based masked face recognition system built on a manually implemented ResNet50 backbone. The key contributions are:

1. A jointly trained multi-task model combining identity embedding via ArcFace and auxiliary mask detection in a single end-to-end architecture.
2. A carefully designed augmentation policy that restricts synthetic occlusion to the eye-level region, preserving lower-face identity cues for same-condition recognition.
3. A gallery-query evaluation protocol that reflects realistic open-set deployment scenarios.
4. A cross-condition experiment (unmasked gallery, masked query) and a threshold calibration analysis providing actionable insights for deployment.

The final system achieves 90.26% identity accuracy and 95.37% mask F1 under same-condition validation. Under the more challenging cross-condition setting, it achieves 52.79% identity accuracy and 99.89% mask F1 at the optimised threshold of 0.38, demonstrating that the model learns meaningful periocular features for masked recognition.

## 6.2 Limitations

- The RMFRD validation evaluation uses identities seen during training; a fully held-out test split on unseen identities would provide a more rigorous upper bound on generalisation.
- Cross-condition performance is substantially lower than same-condition performance, indicating that the representation gap between masked and unmasked embeddings has not been fully bridged.
- Evaluation on a custom real-world dataset with known identities was planned

but not completed.

- The pipeline was not migrated to TPU execution due to CUDA-specific dependencies.

## 6.3 Future Work

1. **Proper train/val/test split**: create a held-out test set of unseen identities for unbiased evaluation.
2. **Custom dataset evaluation**: collect or curate real-world images of known individuals and evaluate the gallery-matching pipeline on them.
3. **Cross-condition training objective**: explicitly train with unmasked-gallery/masked-query pairs using a contrastive or triplet loss to directly minimise the embedding gap between conditions.
4. **Systematic threshold calibration**: perform calibration on a held-out validation set that mirrors the target deployment condition, and report precision–recall curves across the full threshold range.
5. **Larger backbone**: experiment with IResNet100 or ViT-based backbones to assess whether a larger representation capacity reduces the cross-condition gap.
6. **Mobile deployment**: optimise the ONNX model for edge inference using quantisation and pruning.

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